```
================================================================================
EXPERIMENT 7 — ORIGINAL EXP 4 ROTATED 270 DEGREES
================================================================================
Original obstacle count: 57
Rotated obstacle count: 57
Start: (0, 0)
Goal: (19, 19)

Environment verification
------------------------
Grid: 20 x 20
Obstacles: 57
Shortest valid path: 38
Reachable states: 343

================================================================================
EXPERIMENT 7 TRAINING STARTED
================================================================================

Training: Sparse Q-Learning
 Seed 1/10 completed in 14.1 sec
 Seed 2/10 completed in 9.8 sec
 Seed 3/10 completed in 8.7 sec
 Seed 4/10 completed in 8.5 sec
 Seed 5/10 completed in 8.1 sec
 Seed 6/10 completed in 8.4 sec
 Seed 7/10 completed in 6.3 sec
 Seed 8/10 completed in 10.0 sec
 Seed 9/10 completed in 6.1 sec
 Seed 10/10 completed in 8.5 sec
Sparse Q-Learning completed in 88.6 seconds

Training: Fixed PBRS Manhattan
 Seed 1/10 completed in 68.5 sec
 Seed 2/10 completed in 58.5 sec
 Seed 3/10 completed in 62.8 sec
 Seed 4/10 completed in 59.1 sec
 Seed 5/10 completed in 74.6 sec
 Seed 6/10 completed in 58.5 sec
 Seed 7/10 completed in 58.9 sec
 Seed 8/10 completed in 59.9 sec
 Seed 9/10 completed in 58.5 sec
 Seed 10/10 completed in 65.6 sec
Fixed PBRS Manhattan completed in 624.9 seconds

Training: Adaptive PBRS Manhattan
 Seed 1/10 completed in 59.9 sec
 Seed 2/10 completed in 59.8 sec
 Seed 3/10 completed in 61.5 sec
 Seed 4/10 completed in 62.8 sec
 Seed 5/10 completed in 11.4 sec
 Seed 6/10 completed in 61.5 sec
 Seed 7/10 completed in 59.6 sec
 Seed 8/10 completed in 16.3 sec
 Seed 9/10 completed in 58.6 sec
 Seed 10/10 completed in 5.6 sec
Adaptive PBRS Manhattan completed in 456.9 seconds

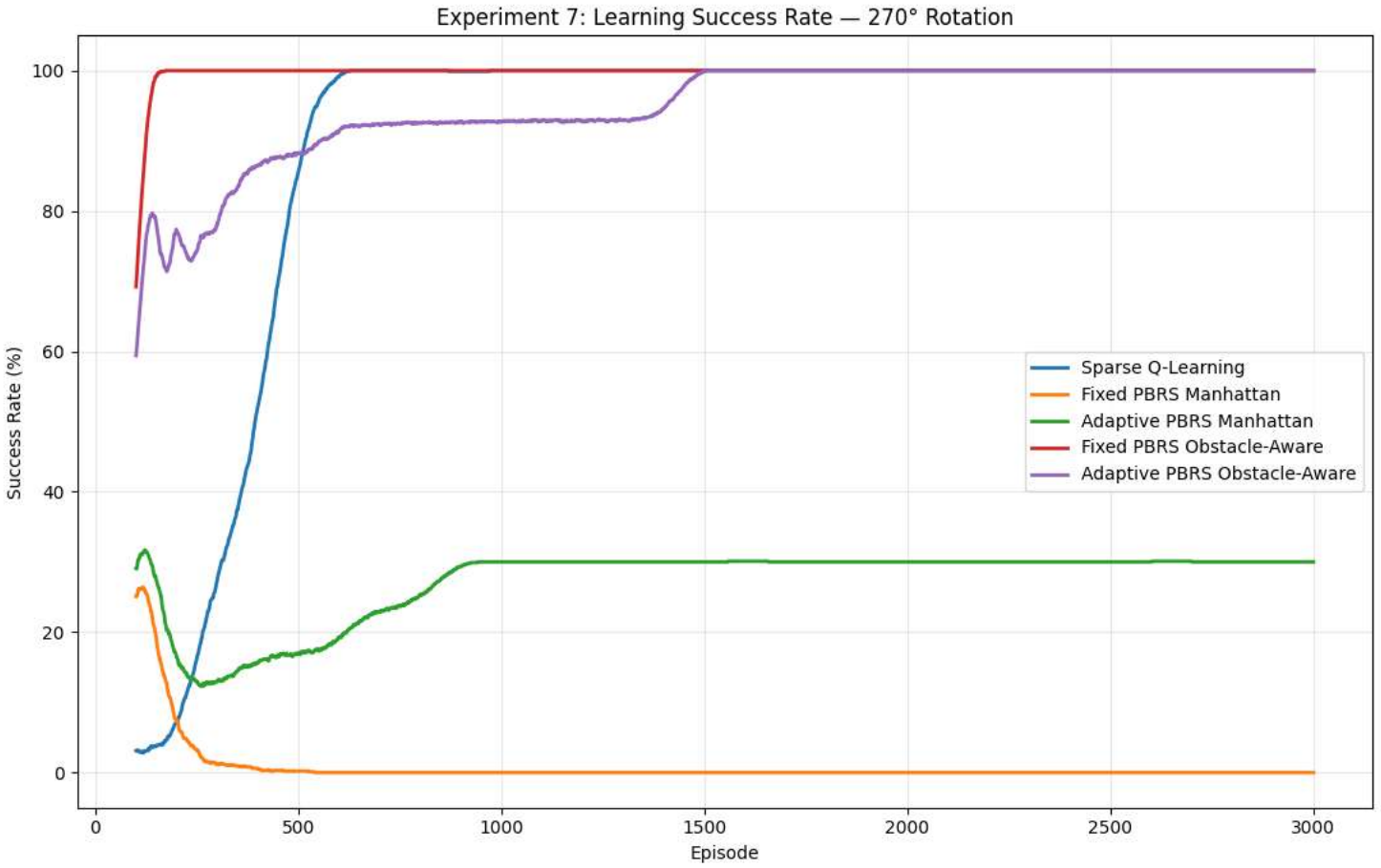
Training: Fixed PBRS Obstacle-Aware
 Seed 1/10 completed in 5.5 sec
 Seed 2/10 completed in 4.4 sec
 Seed 3/10 completed in 4.7 sec
 Seed 4/10 completed in 5.1 sec
 Seed 5/10 completed in 4.4 sec
 Seed 6/10 completed in 5.3 sec
 Seed 7/10 completed in 4.6 sec
 Seed 8/10 completed in 4.3 sec
 Seed 9/10 completed in 5.6 sec
 Seed 10/10 completed in 4.4 sec
Fixed PBRS Obstacle-Aware completed in 48.4 seconds

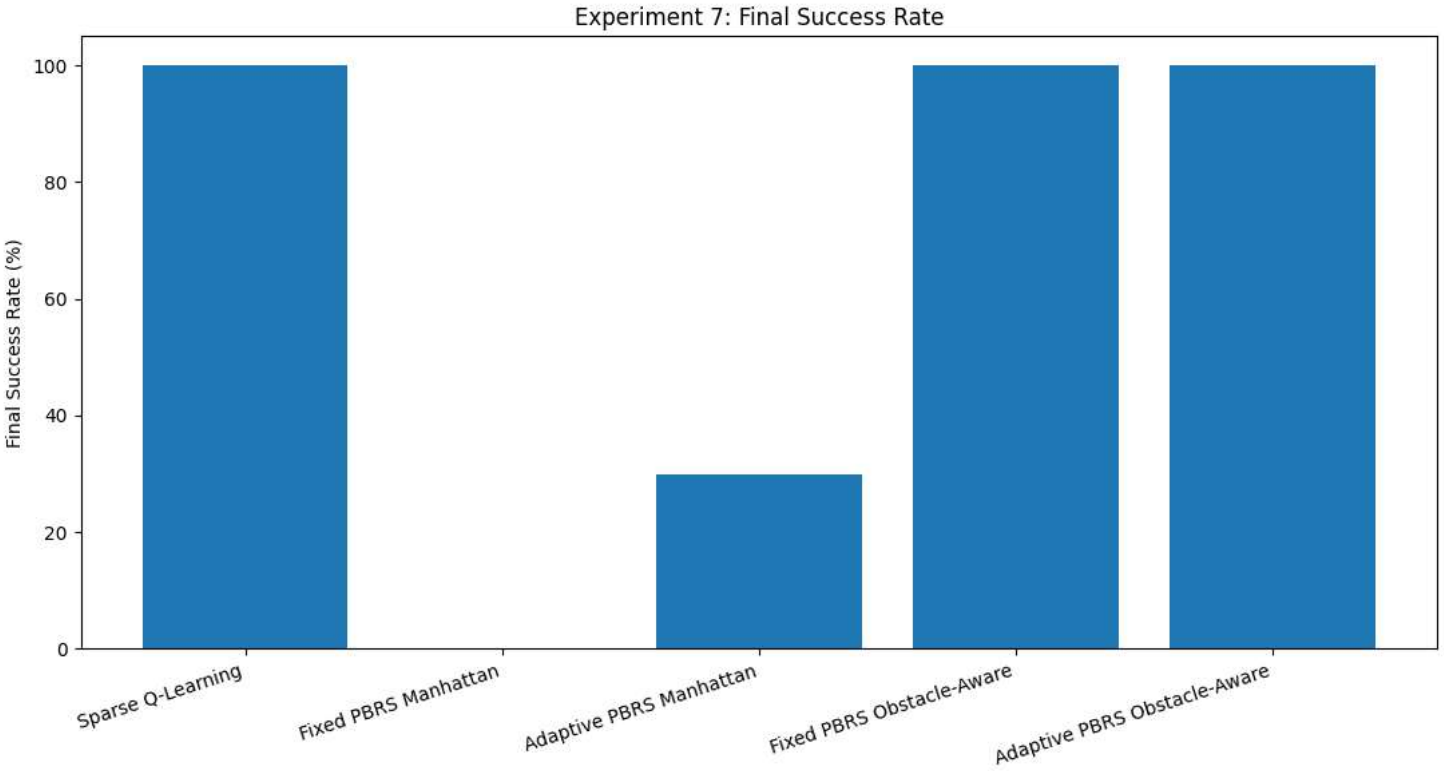
Training: Adaptive PBRS Obstacle-Aware
 Seed 1/10 completed in 5.6 sec
 Seed 2/10 completed in 24.4 sec
 Seed 3/10 completed in 6.2 sec
 Seed 4/10 completed in 4.9 sec
 Seed 5/10 completed in 6.5 sec
 Seed 6/10 completed in 5.5 sec
 Seed 7/10 completed in 9.9 sec
 Seed 8/10 completed in 4.9 sec
 Seed 9/10 completed in 5.5 sec
 Seed 10/10 completed in 5.5 sec
Adaptive PBRS Obstacle-Aware completed in 79.0 seconds

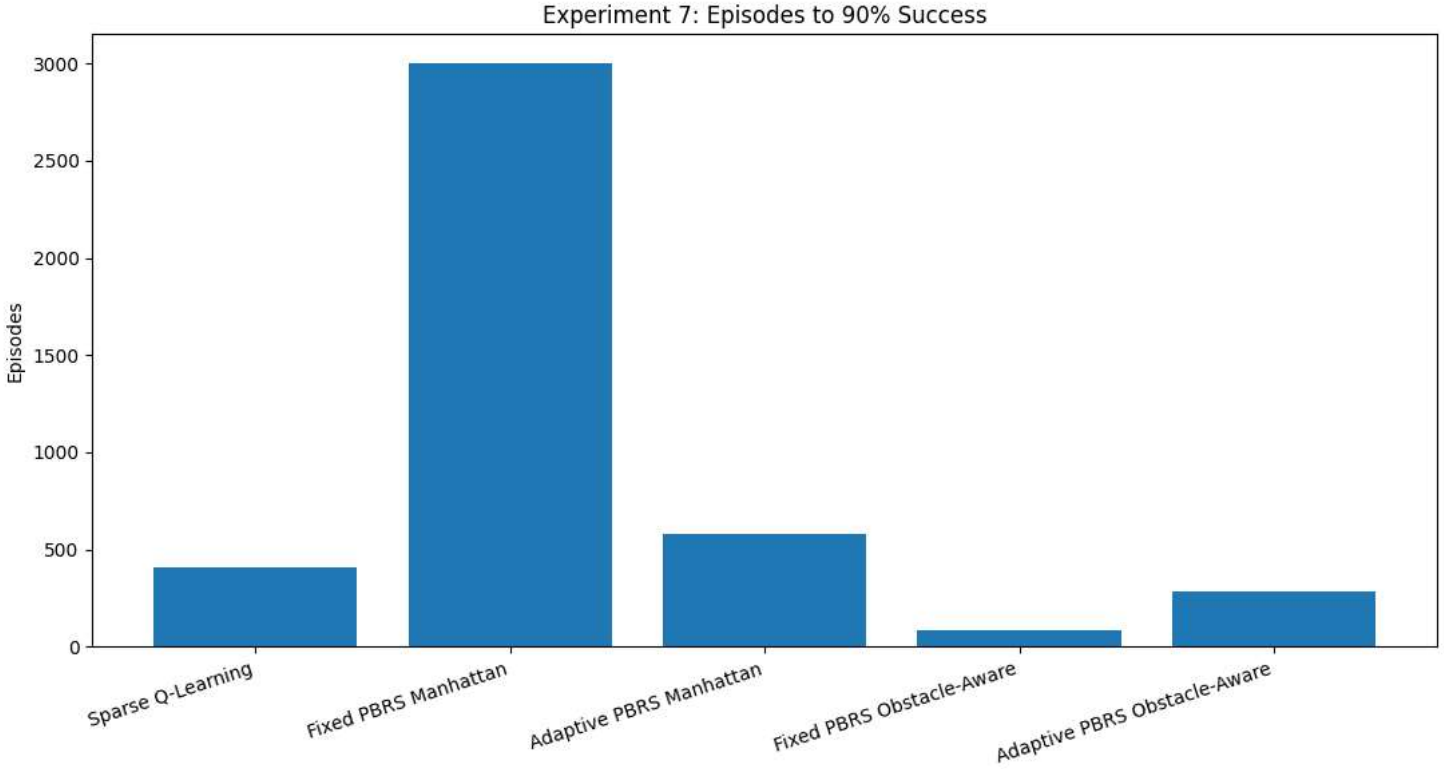
================================================================================================
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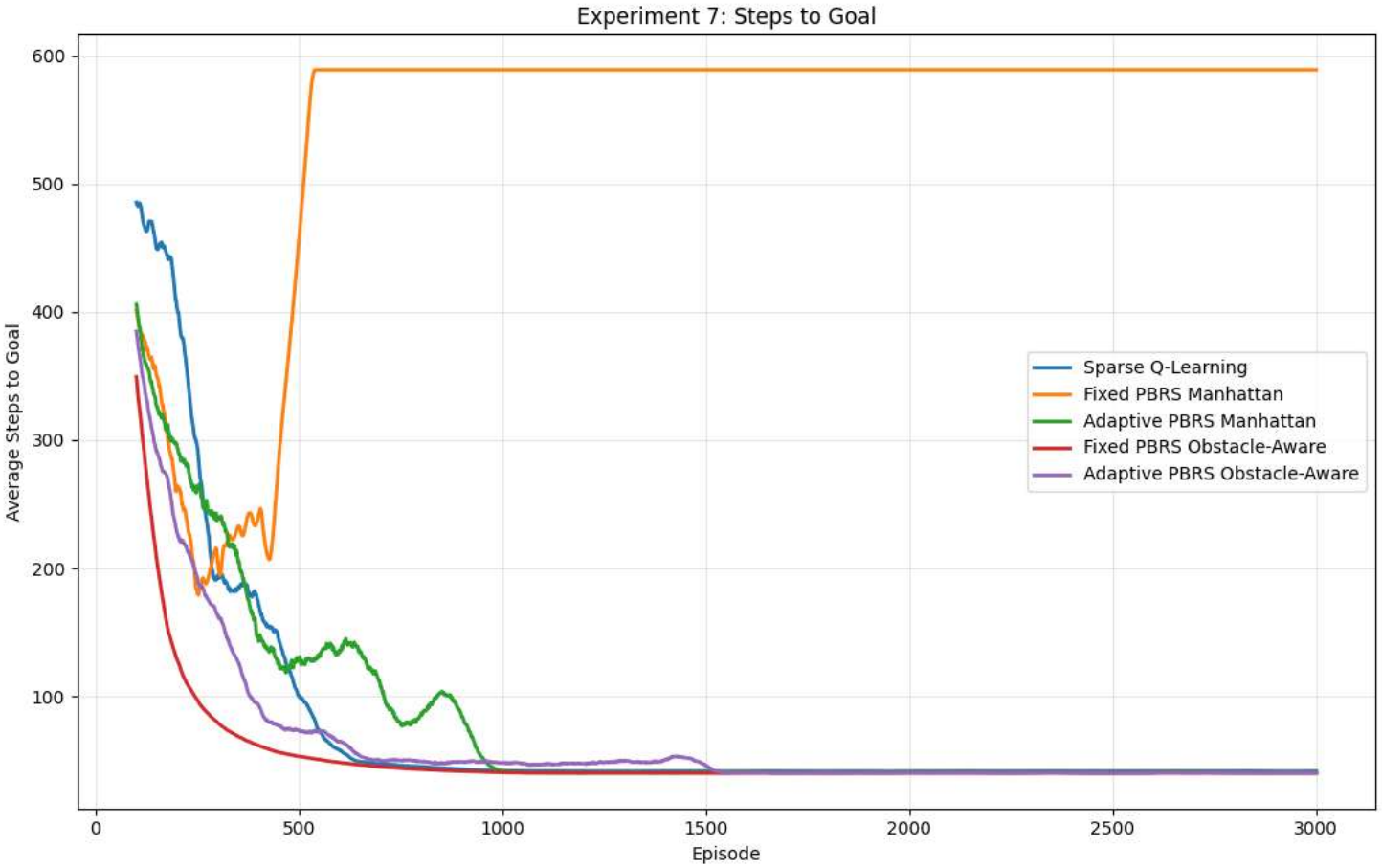
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EXPERIMENT 7 — FINAL RESULTS
================================================================================================
```

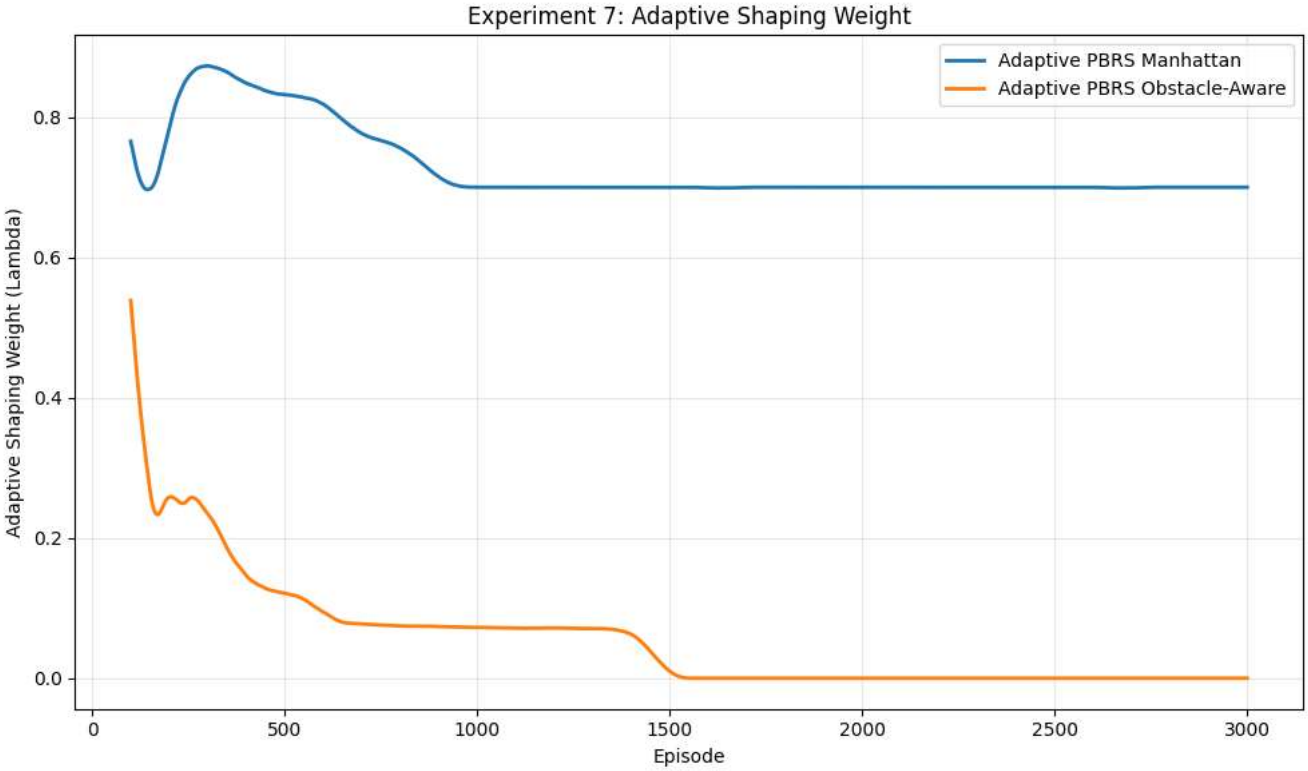
	Method	Final Success Rate (%)	Success SD	Average Steps to Goal	First Successful Episode	Seeds Reaching 90%	Episodes to 90% Success
0	Sparse Q-Learning	100.0	0.0	41.51	20.1	10	406.8
1	Fixed PBRS Manhattan	0.0	0.0	NaN	7.8	0	NaN
2	Adaptive PBRS Manhattan	30.0	48.3	40.25	7.8	3	581.0
3	Fixed PBRS Obstacle-Aware	100.0	0.0	40.17	9.0	10	82.3
4	Adaptive PBRS Obstacle-Aware	100.0	0.0	40.30	9.0	10	286.2

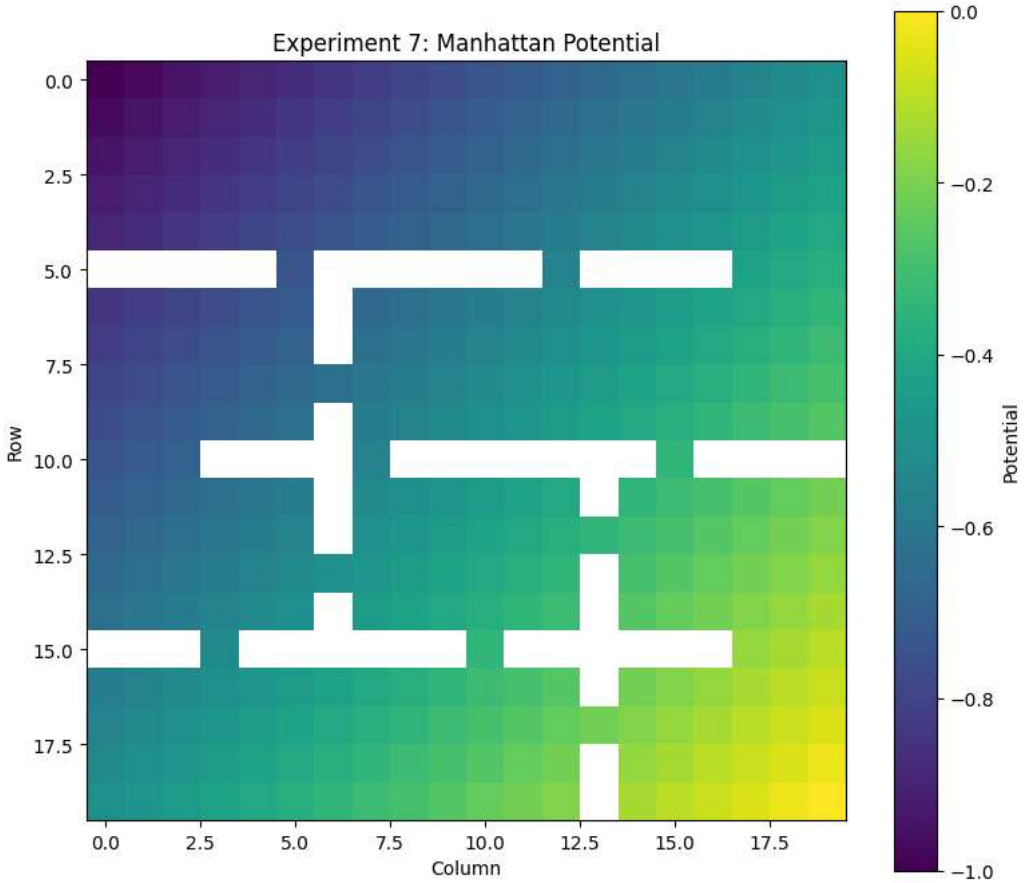


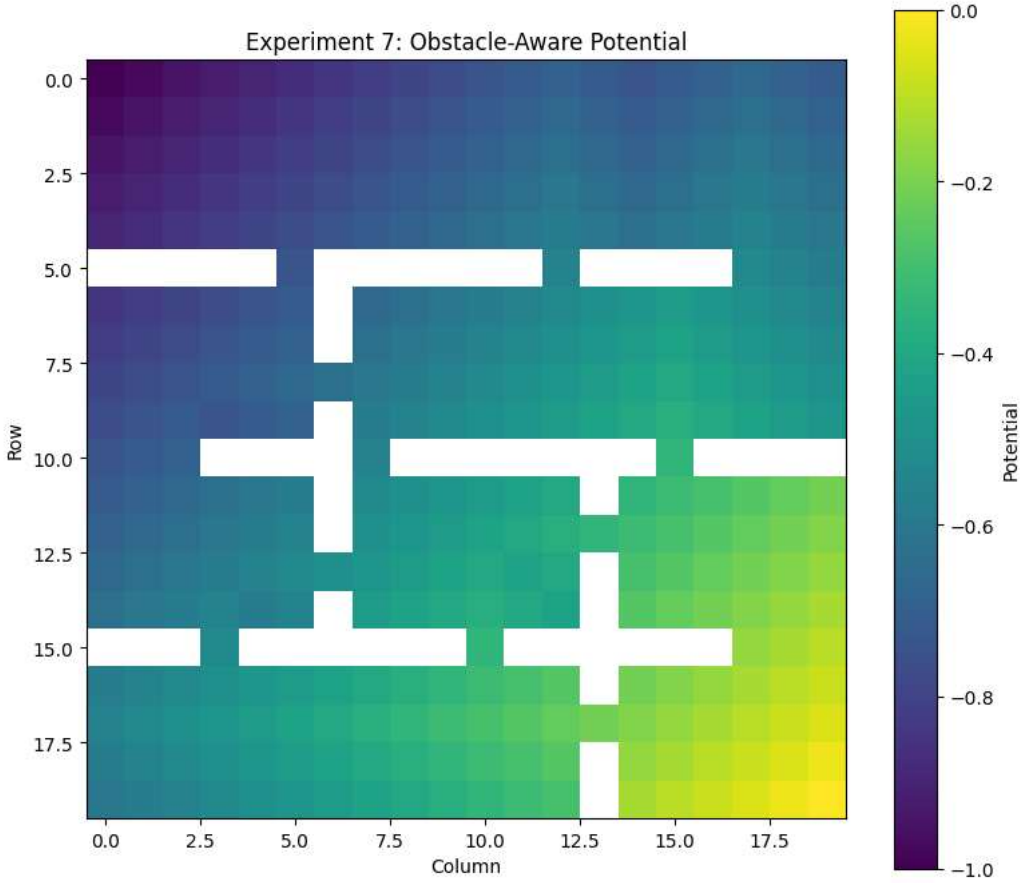

Experiment 7: Final Success Rate
Final Success Rate (%)
0
20
40
60
80
100
Sparse Q-Learning
Fixed PBRS Manhattan
Adaptive PBRS Manhattan
Fixed PBRS Obstacle-Aware
Adaptive PBRS Obstacle-Aware

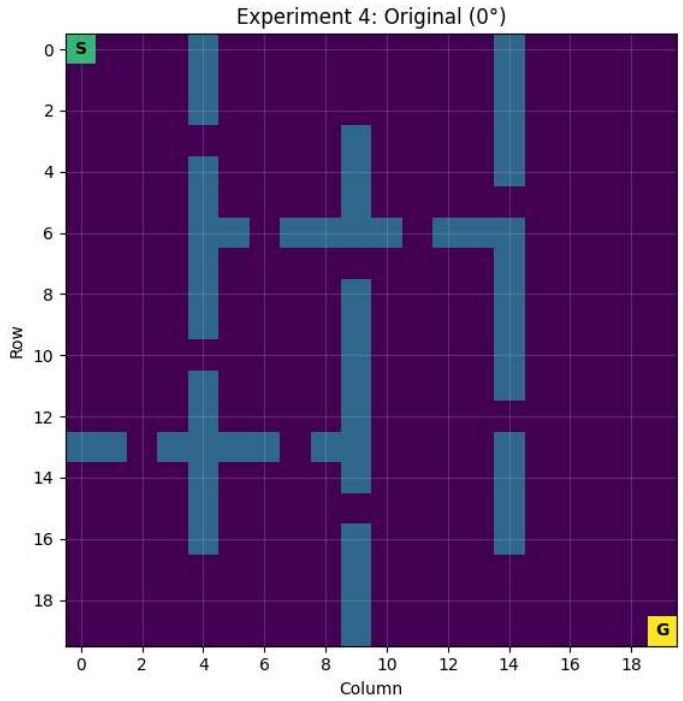

Experiment 7: Episodes to 90% Success
Episodes
0
500
1000
1500
2000
2500
3000
Sparse Q-Learning
Fixed PBRS Manhattan
Adaptive PBRS Manhattan
Fixed PBRS Obstacle-Aware
Adaptive PBRS Obstacle-Aware

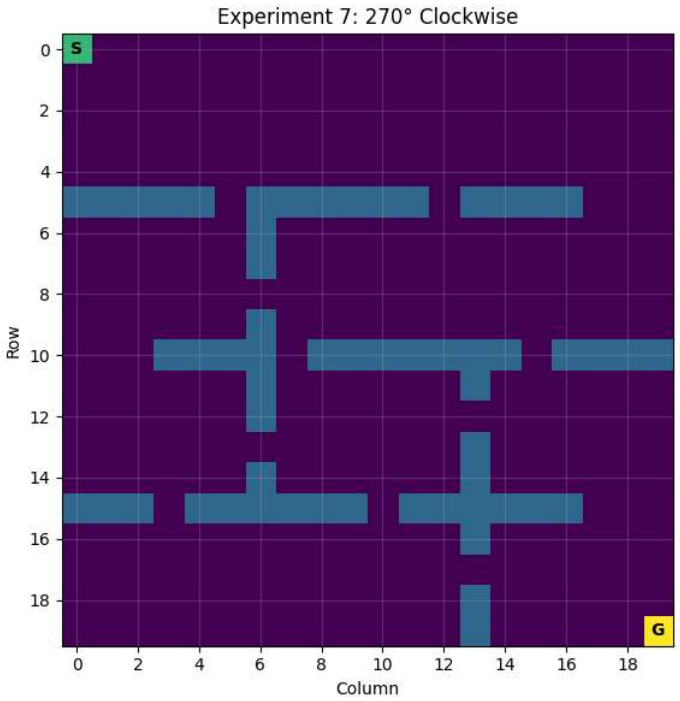

Experiment 7: Steps to Goal
Average Steps to Goal
Episode
Sparse Q-Learning
Fixed PBRS Manhattan
Adaptive PBRS Manhattan
Fixed PBRS Obstacle-Aware
Adaptive PBRS Obstacle-Aware
600
500
400
300
200
100
0
500
1000
1500
2000
2500
3000


Experiment 7: Adaptive Shaping Weight
Adaptive Shaping Weight (Lambda)
Episode
Adaptive PBRS Manhattan
Adaptive PBRS Obstacle-Aware
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
2000
2500
3000


Experiment 7: Manhattan Potential
Row
Column
Potential
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
−0.2
−0.4
−0.6
−0.8
−1.0


Experiment 7: Obstacle-Aware Potential
Row
Column
Potential
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
−0.2
−0.4
−0.6
−0.8
−1.0






```
================================================================================================
EXPERIMENT 7 COMPLETED
================================================================================================
Rotation: 270 degrees clockwise from ORIGINAL Experiment 4
Training episodes: 3000
Seeds: 10
Obstacles: 57
Shortest path: 38
Total execution time: 21.77 minutes

FINAL TABLE:

                      Method  Final Success Rate (%)  Success SD  Average Steps to Goal  First Successful Episode  Seeds Reaching 90%  Episodes
          Sparse Q-Learning                   100.0         0.0                  41.51                      20.1                  10
       Fixed PBRS Manhattan                     0.0         0.0                    NaN                       7.8                   0
    Adaptive PBRS Manhattan                    30.0        48.3                  40.25                       7.8                   3
   Fixed PBRS Obstacle-Aware                   100.0         0.0                  40.17                       9.0                  10
Adaptive PBRS Obstacle-Aware                   100.0         0.0                  40.30                       9.0                  10

Generated files:
1. experiment7_summary.csv
2. experiment7_seed_results.csv
3. exp7_success_rate.png
4. exp7_final_success.png
5. exp7_convergence.png
6. exp7_steps.png
7. exp7_lambda.png
8. exp7_manhattan_potential.png
9. exp7_obstacle_potential.png
10. exp7_environment_comparison.png
```